

ER-EPD075-2

E-Paper Display Module Datasheet



EastRising Technology Co., Limited

Attention:

- A. Some specifications of IC are not listed in this datasheet. Please refer to the IC datasheet for more details.
- B. The related documents for interfacing, demo code, IC datasheet are all available, please download from our web.
- C. Please pay more attention to "INSPECTION CRITERIA" in this datasheet. We assume you already agree with these criterions when you place an order with us. No more recommendations.

REV	Description	Release Date
1.0	Preliminary Release	Jul-09-2020

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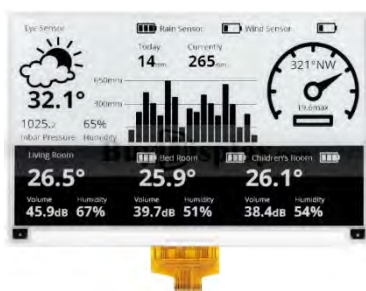
1. ORDERING INFORMATION

1.1 Order Number

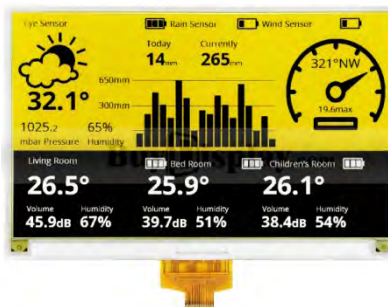
Order Number	Description
ER-EPD075-2B	7.5 inch E-Paper (E-ink) Display White/Black Color
ER-EPD075-2Y	7.5 inch E-Paper (E-ink) Display Yellow/White/Black Color
ER-EPD075-2R	7.5 inch E-Paper (E-ink) Display Red/White/Black Color
ER-EPD075-2-5070	7.5 inch E-Paper (E-ink) Display with Arduino Shield
ER-EPD075-2-5103	7.5 inch E-Paper (E-ink) Display with Raspberry Pi HAT

1.2 Image

ER-EPD075-2B ↓



ER-EPD075-2Y ↓



ER-EPD075-2R ↓



ER-EPD075-2B-5070 ↓



ER-EPD075-2Y-5070 ↓



ER-EPD075-2R-5070 ↓



ER-EPD075-2B-5103 ↓



ER-EPD075-2Y-5103 ↓



ER-EPD075-2R-5103 ↓



2. SPECIFICATION

2.1 Display Specification

Item	Standard Value		Unit
Display Format	880x528		Pixels
Display Connector	FFC		--
FPC Connector	24 Pin,0.5mm Pitch, SMD Horizontal Type Top contact		--
Operating Temperature	ER-EPD075-2B	0 ~ 40	°C
	ER-EPD075-2R	0 ~ 40	
	ER-EPD075-2Y	0 ~ 40	
Storage Temperature	ER-EPD075-2B	0 ~ 40	°C
	ER-EPD075-2R	0 ~ 40	
	ER-EPD075-2Y	0 ~ 40	
Sunlight Readable	Yes		--

2.2 Mechanical Specification

Item	Standard Value	Unit
Screen Size	7.5	inch
Outline Dimension with FPC Folded	170.20(W) x 111.20(H)x1.25(T)	mm
Active Area	163.24(W) x 97.94(H)	mm
Dot Pitch	0.185x 0.185	mm

2.3 Electrical Specification

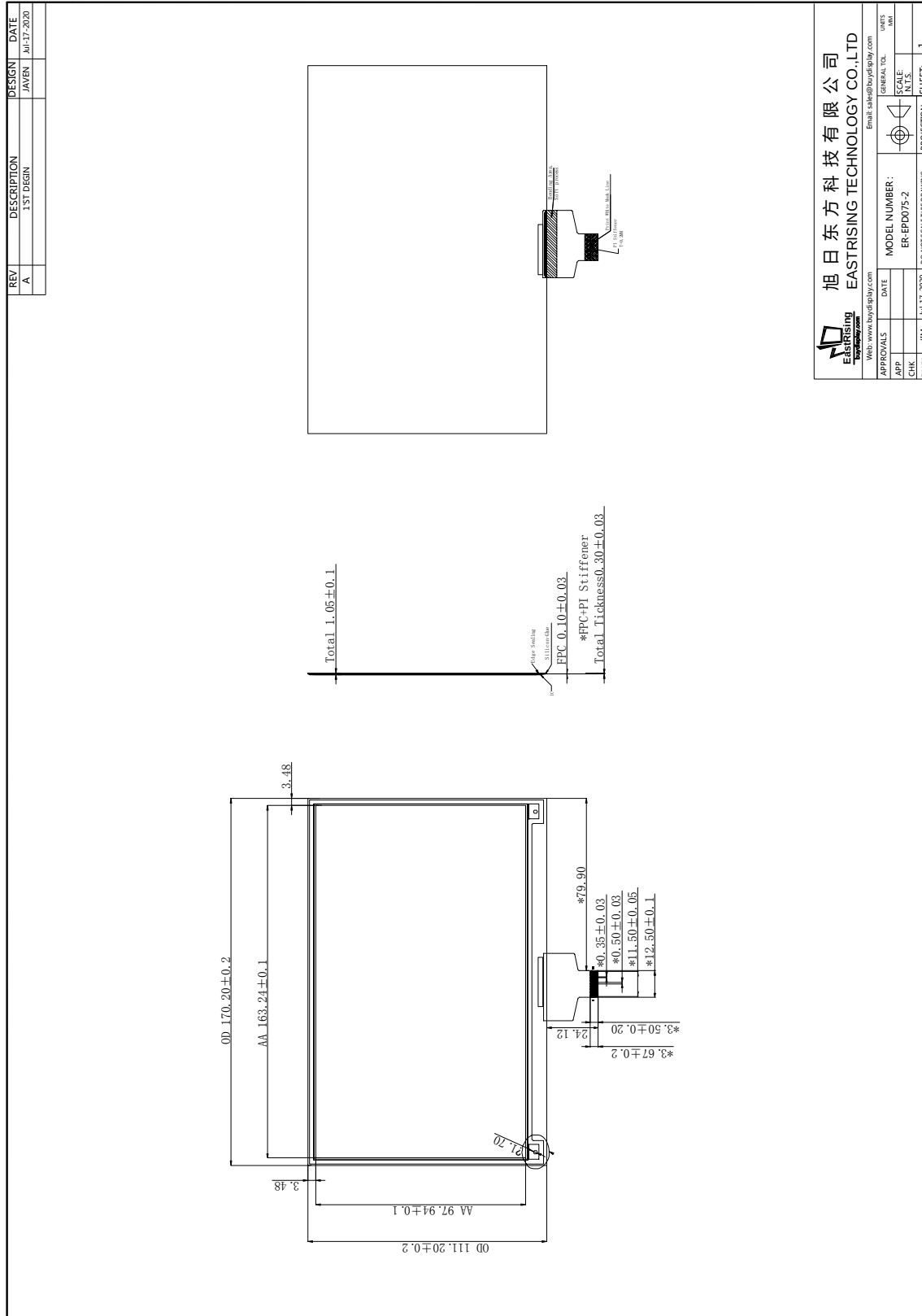
Item	Standard Value	Unit
IC Package	COG	--
Controller	SSD1677	--
Interface	3/4 Wire SPI	--

2.4 Optical Specification

Item	Standard Value	Unit
LCD Type	E-Ink Display (E-Paper Display)	--
Viewing Angle Range	Left:85 , Right:85 , Up:85 , Down:85	deg

3. OUTLINE DRAWING

3.1 ER-EPD075-2 Outline Drawing



4. ELECTRICAL SPEC

4.1 Pin Configuration

Pin #	Single	Description	Remark
1	NC	No connection and do not connect with other NC pins	Keep Open
2	GDR	N-Channel MOSFET Gate Drive Control	
3	RESE	Current Sense Input for the Control Loop	
4	NC	No connection and do not connect with other NC pins e	Keep Open
5	VSH2	Positive Source driving voltage	
6	TSCCL	I2C Interface to digital temperature sensor Clock pin	
7	TSDA	I2C Interface to digital temperature sensor Date pin	
8	BS1	Bus selection pin	Note 4-5
9	BUSY	Busy state output pin	Note 4-4
10	RES #	Reset	Note 4-3
11	D/C #	Data /Command control pin	Note 4-2
12	CS #	Chip Select input pin	Note 4-1
13	SCL	serial clock pin (SPI)	
14	SDA	serial data pin (SPI)	
15	VDDIO	Power for interface logic pins	
16	VCI	Power Supply pin for the chip	
17	VSS	Ground	
18	VDD	Core logic power pin	
19	VPP	Power Supply for OTP Programming	
20	VSH1	Positive Source driving voltage	
21	VGH	Power Supply pin for Positive Gate driving voltage and VSH	
22	VSL	Negative Source driving voltage	
23	VGL	Power Supply pin for Negative Gate driving voltage, VCOM and VSL	
24	VCOM	VCOM driving voltage	

Note 4-1: This pin (CS#) is the chip select input connecting to the MCU. The chip is enabled for MCU communication: only when CS# is pulled LOW.

Note 4-2: This pin (D/C#) is Data/Command control pin connecting to the MCU. When the pin is pulled HIGH, the data will be interpreted as data. When the pin is pulled LOW, the data will be interpreted as command.

Note 4-3: This pin (RES#) is reset signal input. The Reset is active low.

Note 4-4: This pin (BUSY) is Busy state output pin. When Busy is High the operation of chip should not be interrupted and any commands should not be issued to the module. The driver IC will put Busy pin High when the driver IC is working such as:

- Outputting display waveform; or
- Communicating with digital temperature sensor

Note 4-5: This pin (BS1) is for 3-line SPI or 4-line SPI selection. When it is “Low”, 4-line SPI is selected. When it is “High”, 3-line SPI (9 bits SPI) is selected.

5. MCU Interface

5.1 MCU interface selection

ER-EPD075-2 can support 4-wire or 3-wire serial peripheral MCU interface, which is pin selectable by BS1 pin. The interface pin assignment for different MCU interfaces is shown in Table 5-1.

Note

- (1) L is connected to VSS
- (2) H is connected to VDDIO

Table 5-1: Interface pin assignment for different MCU interfaces

MCU Interface	Pin Name						
	BS1	RES#	CS#	D/C#	SCL	SDI	SDO
4-wire serial peripheral interface (SPI)	L	Required	Required	Required	SCL	SDI	SDO
3-wire serial peripheral interface (SPI) – 9 bits SPI	H	Required	Required	L	SCL	SDI	SDO

5.2 MCU Serial Peripheral Interface (4-wire SPI)

The 4-wire SPI consists of serial clock SCL, serial data input SDI, D/C# and CS#. The control pins status in

4-wire SPI in writing command/data is shown in Table 5-2 and the write procedure in 4-wire SPI is shown in

Figure 5-1

Table 5-2 : Control pins status of 4-wire SPI

Function	SCL pin	SDI pin	D/C# pin	CS# pin
Write command	↑	Command bit	L	L
Write data	↑	Data bit	H	L

Note:

- (1) L is connected to VSS and H is connected to VDDIO
- (2) ↑ stands for rising edge of signal
- (3) SDI is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0. The level of D/C# should be kept over the whole byte. The data byte in the shift register is written to the Graphic Display Data RAM (RAM)/Data Byte register or command Byte register according to D/C# pin.

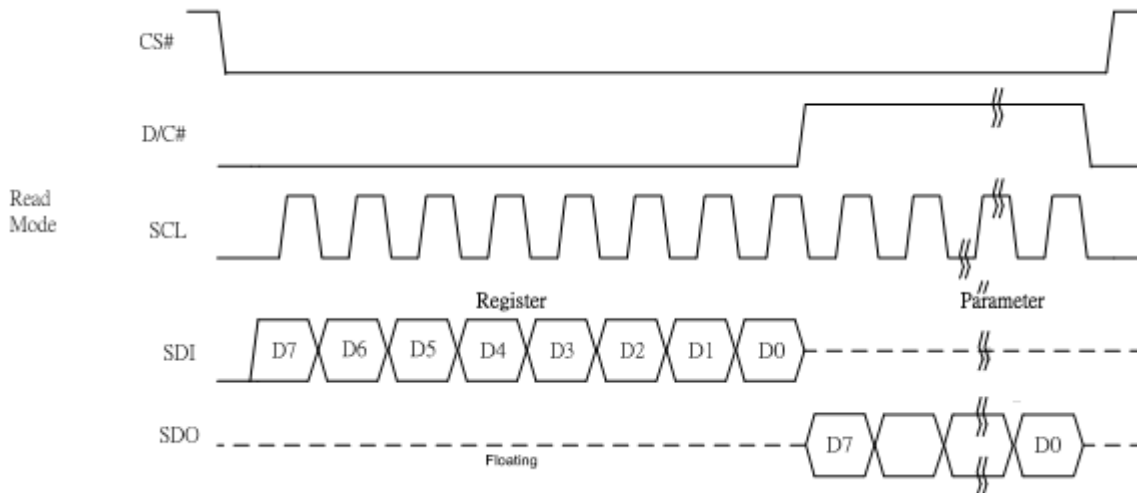


Figure 5-1: wire mode

In the read operation, after CS# is pulled low, the first byte sent is command byte, D/C# is pulled low. After command byte sent, the following byte(s) read are data byte(s), so D/C# bit is then pulled high. An 8-bit data will be shifted out on every clock falling edge. The serial data output SDO bit shifting sequence is D7, D6, to D0 bit. Figure 5-2 shows the read procedure in 4-wire SPI mode.

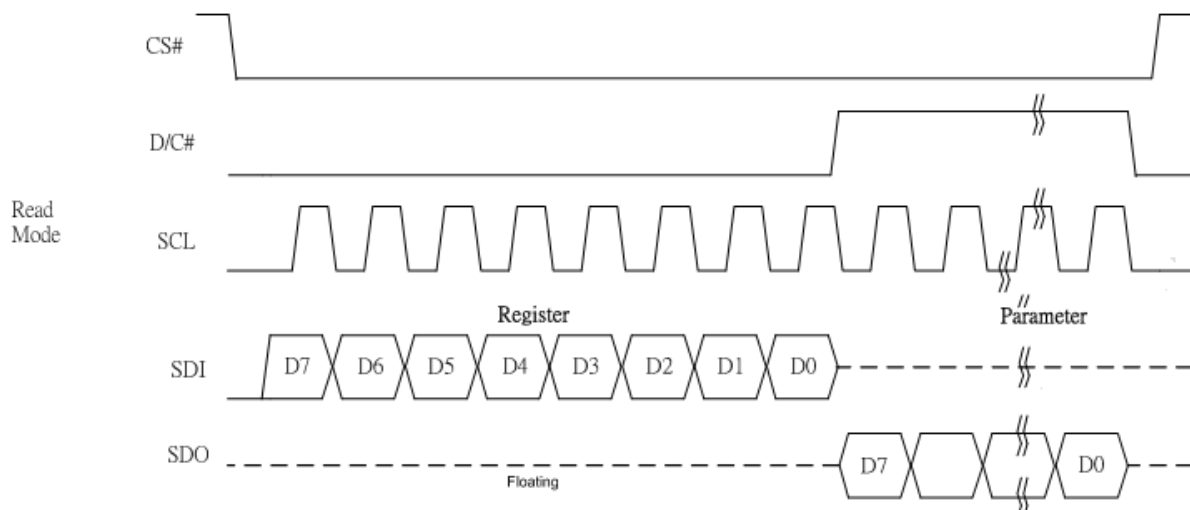


Figure 5-2: Read procedure in 4-wire SPI mode

5.3 MCU Serial Peripheral Interface (3-wire SPI)

The 3-wire SPI consists of serial clock SCL, serial data input SDI, and CS#. The operation is similar to 4-wire SPI while D/C# pin is not used and it must be tied to LOW. The control pins status in 3-wire SPI is shown in Table 5-3.

In the write operation, a 9-bit data will be shifted into the shift register on every clock rising edge. The bit shifting sequence is D/C# bit, D7 bit, D6 bit to D0 bit. The first bit is D/C# bit which determines the following byte is command or data. When D/C# bit is 0, the following byte is command. When D/C# bit is 1, the following byte is data. Table 5-3 shows the write procedure in 3-wire SPI

Table 5-3 : Control pins status of 3-wire SPI

Function	SCL pin	SDI pin	D/C# pin	CS# pin
Write command	↑	Command bit	Tie LOW	L
Write data	↑	Data bit	Tie LOW	L

Note:

- (1) L is connected to V_{SS} and H is connected to V_{DDIO}
- (2) ↑ stands for rising edge of signal

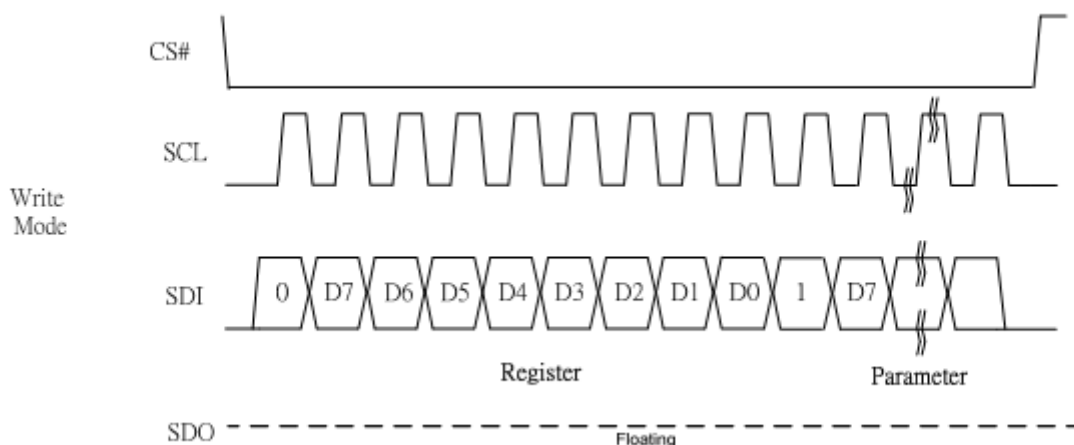


Figure 5-3: Write procedure in 3-wire SPI mode

In the read operation, serial data are transferred in the unit of 9 bits. After CS# pull low, the first byte is command byte, the D/C# bit is as 0 and following with the register byte. After command byte send, the following byte(s) are data byte(s), with D/C# bit is 1. After D/C# bit sending from MCU, an 8-bit data will be shifted out on every clock falling edge. The serial data output SDO bit shifting sequence is D7, D6, to D0 bit. Figure 5-4 shows the read procedure in 3-wire SPI.

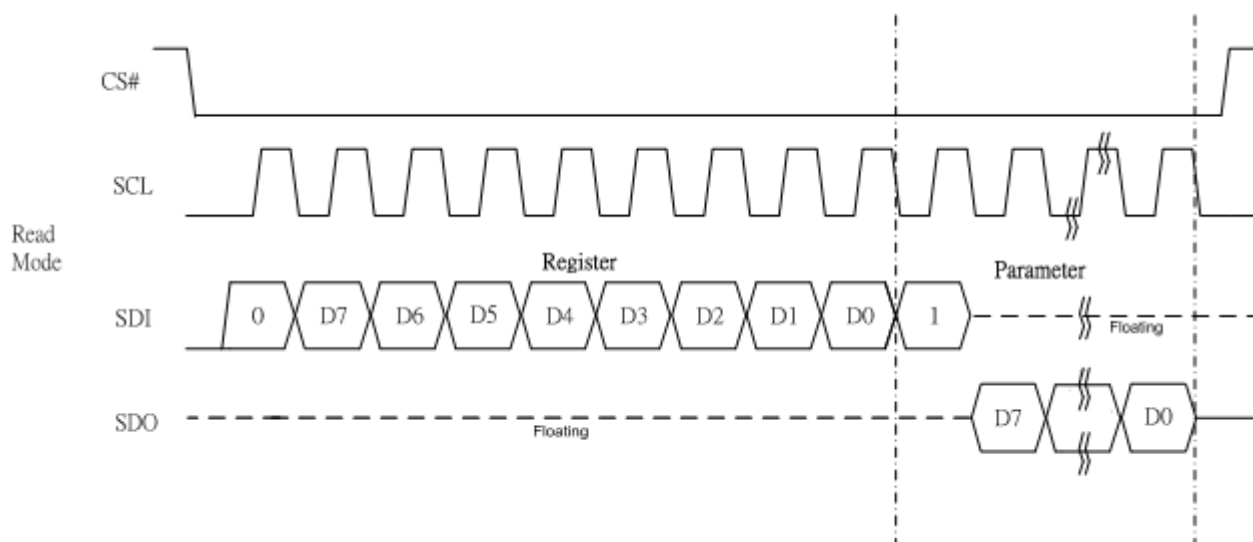


Figure 5-4: Read procedure in 3-wire SPI mode

6. COMMAND TABLE

Command Table																																																								
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																																												
0	0	01	0	0	0	0	0	0	0	1	Driver Output control	Gate setting A[9:0]= 2A7h [POR], 680 MUX MUX Gate lines setting as (A[9:0] + 1).																																												
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																														
0	1		0	0	0	0	0	0	A9	A8																																														
0	1		0		0	0	0	B2	B1	B0																																														
												B[2]: GD Selects the 1st output Gate GD=0 [POR], G0 is the 1st gate output channel, gate output sequence is G0,G1, G2, G3, ... GD=1, G1 is the 1st gate output channel, gate output sequence is G1, G0, G3, G2, ... B[1]: SM Change scanning order of gate driver. SM=0 [POR], G0, G1, G2, G3 ...679 (left and right gate interlaced) SM=1, G0, G2, G4 ...G678, G1, G3, ...G679 B[0]: TB TB = 0 [POR], scan from G0 to G679 TB = 1, scan from G679 to G0.																																												
0	0	03	0	0	0	0	0	0	1	1	Gate Driving voltage Control	Set Gate driving voltage A[4:0] = 00h [POR] VGH setting from 12V to 20V																																												
												<table><tr><th>A[4:0]</th><th>VGH</th><th>A[4:0]</th><th>VGH</th></tr><tr><td>00h</td><td>20</td><td>10h</td><td>16.5</td></tr><tr><td>07h</td><td>12</td><td>11h</td><td>17</td></tr><tr><td>08h</td><td>12.5</td><td>12h</td><td>17.5</td></tr><tr><td>09h</td><td>13</td><td>13h</td><td>18</td></tr><tr><td>0Ah</td><td>13.5</td><td>14h</td><td>18.5</td></tr><tr><td>0Bh</td><td>14</td><td>15h</td><td>19</td></tr><tr><td>0Ch</td><td>14.5</td><td>16h</td><td>19.5</td></tr><tr><td>0Dh</td><td>15</td><td>17h</td><td>20</td></tr><tr><td>0Eh</td><td>15.5</td><td>Other</td><td>NA</td></tr><tr><td>0Fh</td><td>16</td><td></td><td></td></tr></table>	A[4:0]	VGH	A[4:0]	VGH	00h	20	10h	16.5	07h	12	11h	17	08h	12.5	12h	17.5	09h	13	13h	18	0Ah	13.5	14h	18.5	0Bh	14	15h	19	0Ch	14.5	16h	19.5	0Dh	15	17h	20	0Eh	15.5	Other	NA	0Fh	16		
A[4:0]	VGH	A[4:0]	VGH																																																					
00h	20	10h	16.5																																																					
07h	12	11h	17																																																					
08h	12.5	12h	17.5																																																					
09h	13	13h	18																																																					
0Ah	13.5	14h	18.5																																																					
0Bh	14	15h	19																																																					
0Ch	14.5	16h	19.5																																																					
0Dh	15	17h	20																																																					
0Eh	15.5	Other	NA																																																					
0Fh	16																																																							

Command Table												
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	04	0	0	0	0	0	1	0	0	Source Driving voltage Control	Set Source driving voltage A[7:0] = 41h [POR], VSH1 at 15V B[7:0] = A8h [POR], VSH2 at 5V. C[7:0] = 32h [POR], VSL at -15V
0	1		A7	A6	A5	A4	A3	A2	A1	A2		
0	1		B7	B6	B5	B4	B3	B2	B1	B0		
0	1		C7	C6	C5	C4	C3	C2	C1	C0		
B[7] = 1, VSH2 voltage setting from 2.4V to 8.8V			A[7]/B[7] = 0, VSH1/VSH2 voltage setting from 9V to 17V					C[7] = 0, VSL setting from -9V to -17V				
A/B[7:0]			VSH1/VSH2		A/B[7:0]		VSH1/VSH2		A/B[7:0]		VSH1/VSH2	
8Eh			2.4		AFh		5.7		23h		9	
8Fh			2.5		B0h		5.8		24h		9.2	
90h			2.6		B1h		5.9		25h		9.4	
91h			2.7		B2h		6		26h		9.6	
92h			2.8		B3h		6.1		27h		9.8	
93h			2.9		B4h		6.2		28h		10	
94h			3		B5h		6.3		29h		10.2	
95h			3.1		B6h		6.4		2Ah		10.4	
96h			3.2		B7h		6.5		2Bh		10.6	
97h			3.3		B8h		6.6		2Ch		10.8	
98h			3.4		B9h		6.7		2Dh		11	
99h			3.5		BAh		6.8		2Eh		11.2	
9Ah			3.6		BBh		6.9		2Fh		11.4	
9Bh			3.7		BCh		7		30h		11.6	
9Ch			3.8		BDh		7.1		31h		11.8	
9Dh			3.9		BEh		7.2		32h		12	
9Eh			4		BFh		7.3		33h		12.2	
9Fh			4.1		C0h		7.4		34h		12.4	
A0h			4.2		C1h		7.5		35h		12.6	
A1h			4.3		C2h		7.6		36h		12.8	
A2h			4.4		C3h		7.7		37h		13	
A3h			4.5		C4h		7.8		38h		13.2	
A4h			4.6		C5h		7.9		39h		13.4	
A5h			4.7		C6h		8		3Ah		13.6	
A6h			4.8		C7h		8.1		3Bh		13.8	
A7h			4.9		C8h		8.2					
A8h			5		C9h		8.3					
A9h			5.1		CAh		8.4					
AAh			5.2		CBh		8.5					
ABh			5.3		CCh		8.6					
ACh			5.4		CDh		8.7					
ADh			5.5		CEh		8.8					
AEh			5.6		Other		NA					
Remark: VSH1> VSH2												
0	0	0F	0	0	0	0	1	1	1	1	Gate scan start position	Set the scanning start position of the gate driver. The valid range is from 0 to 679. A[9:0] = 000h [POR] When TB=0: SCN [9:0] = A[9:0] When TB=1: SCN [9:0] = 679 - A[9:0]
0	1		A7	A6	A5	A4	A3	A2	A1	A0		
0	1		0	0	0	0	0	0	A9	A8		

Command Table													
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description	
0	0	10	0	0	0	1	0	0	0	0	Deep Sleep mode	Deep Sleep mode Control:	
0	0		0	0	0	0	0	A2	A1	A0		A[1:0] :	Description
												00	Normal Mode [POR]
												01	Enter Deep Sleep Mode 1
											11	Enter Deep Sleep Mode 2	
												After this command initiated, the chip will enter Deep Sleep Mode, BUSY pad will keep output high. Remark: To Exit Deep Sleep mode, User required to send HWRESET to the driver	
0	0	11	0	0	0	1	0	0	0	1	Data mode setting	Entry	Define data entry sequence
0	1		0	0	0	0	0	A2	A1	A0			A[2:0] = 011 [POR] A [1:0] = ID[1:0] Address automatic increment / decrement setting The setting of incrementing or decrementing of the address counter can be made independently in each upper and lower bit of the address. 00 –Y decrement, X decrement, 01 –Y decrement, X increment, 10 –Y increment, X decrement, 11 –Y increment, X increment [POR] A[2] = AM Set the direction in which the address counter is updated automatically after data are written to the RAM. AM= 0, the address counter is updated in the X direction. [POR] AM = 1, the address counter is updated in the Y direction.
0	0	12	0	0	0	1	0	0	1	0	SW RESET	It resets the commands and parameters to their S/W Reset default values except R10h-Deep Sleep Mode During operation, BUSY pad will output high. Note: RAM are unaffected by this command.	

Command Table																											
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description															
0	0	14	0	0	0	1	0	1	0	0	HV Ready Detection	HV ready detection A[6:0] = 00h [POR] The command required CLKEN=1 and ANALOGEN=1. Refer to Register 0x22 for detail. After this command initiated, HV Ready detection starts. BUSY pad will output high during detection. The detection result can be read from the Status Bit Read (Command 0x2F).															
0	0	15	0	0	0	1	0	1	0	1	VCI Detection	VCI Detection A[2:0] = 100 [POR] , Detect level at 2.3V A[2:0] : VCI level Detect<table><tr><th>A[2:0]</th><th>VCI level</th></tr><tr><td>011</td><td>2.2V</td></tr><tr><td>100</td><td>2.3V</td></tr><tr><td>101</td><td>2.4V</td></tr><tr><td>110</td><td>2.5V</td></tr><tr><td>111</td><td>2.6V</td></tr><tr><td>Other</td><td>NA</td></tr></table> The command required CLKEN=1 and ANALOGEN=1 Refer to Register 0x22 for detail. After this command initiated, VCI detection starts. BUSY pad will output high during detection. The detection result can be read from the Status Bit Read (Command 0x2F).		A[2:0]	VCI level	011	2.2V	100	2.3V	101	2.4V	110	2.5V	111	2.6V	Other	NA
A[2:0]	VCI level																										
011	2.2V																										
100	2.3V																										
101	2.4V																										
110	2.5V																										
111	2.6V																										
Other	NA																										
0	1		0	0	0	0	0	A2	A1	A0																	
0	0	18	0	0	0	1	1	0	0	0	Temperature Sensor Control	Temperature Sensor Selection A[7:0] = 48h [POR], external temperature sensor A[7:0] = 80h Internal temperature sensor															
0	1		A7	A6	A5	A4	A3	A2	A1	A0																	
0	0	1A	0	0	0	0	0	0	1	0	Temperature Sensor Control (Write to temperature register)	Write to temperature register. A[11:0] = 7FFh[POR]															
0	1		A11	A10	A9	A8	A7	A6	A5	A4																	
0	1		A3	A2	A1	A0	0	0	0	0																	
0	0	1B	0	0	0	1	1	0	1	1	Temperature Sensor Control (Read from temperature register)	Read from temperature register.															
1	1		A11	A10	A9	A8	A7	A6	A5	A4																	
1	1		A3	A2	A1	A0	0	0	0	0																	

Command Table																								
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description												
0	0	1C	0	0	0	1	1	1	0	0	Temperature Sensor Control (Write Command to External temperature sensor)	Write Command to External temperature sensor. A[7:0] = 00h [POR], B[7:0] = 00h [POR], C[7:0] = 00h [POR], A[7:6] <table><tr><td>A[7:6]</td><td>Select no of byte to be sent</td></tr><tr><td>00</td><td>Address + pointer</td></tr><tr><td>01</td><td>Address + pointer + 1st arameter</td></tr><tr><td>10</td><td>Address + pointer + 1st parameter +2nd pointer</td></tr><tr><td>11</td><td>Address</td></tr></table> A[5:0] – Pointer Setting B[7:0] – 1st parameter C[7:0] – 2nd parameter The command required CLKEN=1. Refer to Register 0x22 for detail. After this command initiated, Write Command to external temperature sensor starts. BUSY pad will output high during operation.	A[7:6]	Select no of byte to be sent	00	Address + pointer	01	Address + pointer + 1st arameter	10	Address + pointer + 1st parameter +2nd pointer	11	Address		
A[7:6]	Select no of byte to be sent																							
00	Address + pointer																							
01	Address + pointer + 1st arameter																							
10	Address + pointer + 1st parameter +2nd pointer																							
11	Address																							
0	1		A7	A6	A5	A4	A3	A2	A1	A0														
0	1		B7	B6	B5	B4	B3	B2	B1	B0	temperature sensor)	B[7:0] = 00h [POR], C[7:0] = 00h [POR], A[7:6] <table><tr><td>A[7:6]</td><td>Select no of byte to be sent</td></tr><tr><td>00</td><td>Address + pointer</td></tr><tr><td>01</td><td>Address + pointer + 1st arameter</td></tr><tr><td>10</td><td>Address + pointer + 1st parameter +2nd pointer</td></tr><tr><td>11</td><td>Address</td></tr></table> A[5:0] – Pointer Setting B[7:0] – 1st parameter C[7:0] – 2nd parameter The command required CLKEN=1. Refer to Register 0x22 for detail. After this command initiated, Write Command to external temperature sensor starts. BUSY pad will output high during operation.	A[7:6]	Select no of byte to be sent	00	Address + pointer	01	Address + pointer + 1st arameter	10	Address + pointer + 1st parameter +2nd pointer	11	Address		
A[7:6]	Select no of byte to be sent																							
00	Address + pointer																							
01	Address + pointer + 1st arameter																							
10	Address + pointer + 1st parameter +2nd pointer																							
11	Address																							
0	1		C7	C6	C5	C4	C3	C2	C1	C0														
0	0	20	0	0	1	0	0	0	0	0	Master Activation	Activate Display Update Sequence The Display Update Sequence Option is located at R22h. BUSY pad will output high during operation. User should not interrupt this operation to avoid corruption of panel images.												
0	0	21	0	0	1	0	0	0	0	1	Display Update Control 1	RAM content option for Display Update A[7:0] = 00h [POR] A[7:4] Red RAM option <table><tr><td>0000</td><td>Normal</td></tr><tr><td>0100</td><td>Bypass RAM content as 0</td></tr><tr><td>1000</td><td>Inverse RAM content</td></tr></table> A[3:0] BW RAM option <table><tr><td>0000</td><td>Normal</td></tr><tr><td>0100</td><td>Bypass RAM content as 0</td></tr><tr><td>1000</td><td>Inverse RAM content</td></tr></table>	0000	Normal	0100	Bypass RAM content as 0	1000	Inverse RAM content	0000	Normal	0100	Bypass RAM content as 0	1000	Inverse RAM content
0000	Normal																							
0100	Bypass RAM content as 0																							
1000	Inverse RAM content																							
0000	Normal																							
0100	Bypass RAM content as 0																							
1000	Inverse RAM content																							
0	1		A7	A6	A5	A4	A3	A2	A1	A0														
0	0	22	0	0	1	0	0	0	1	0	Display Update Control 2	Display Update Sequence Option: Enable the stage for Master Activation A[7:0]= FFh (POR) <table><tr><td></td><td>Parameter (in Hex)</td></tr><tr><td>Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 1 Then Disable ANALOG Then Disable OSC</td><td>C7</td></tr><tr><td>Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 2 Then Disable ANALOG Then Disable OSC</td><td>CF</td></tr><tr><td colspan="2"></td></tr><tr><td>Enable Clock Signal, Then Load LUT with DISPLAY Mode 1</td><td>90</td></tr><tr><td>Enable Clock Signal, Then Load Temperature value from I2C Single Master Interface Then Load LUT with DISPLAY Mode 1</td><td>B0</td></tr></table>		Parameter (in Hex)	Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 1 Then Disable ANALOG Then Disable OSC	C7	Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 2 Then Disable ANALOG Then Disable OSC	CF			Enable Clock Signal, Then Load LUT with DISPLAY Mode 1	90	Enable Clock Signal, Then Load Temperature value from I2C Single Master Interface Then Load LUT with DISPLAY Mode 1	B0
	Parameter (in Hex)																							
Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 1 Then Disable ANALOG Then Disable OSC	C7																							
Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 2 Then Disable ANALOG Then Disable OSC	CF																							
Enable Clock Signal, Then Load LUT with DISPLAY Mode 1	90																							
Enable Clock Signal, Then Load Temperature value from I2C Single Master Interface Then Load LUT with DISPLAY Mode 1	B0																							
0	1		A7	A6	A5	A4	A3	A2	A1	A0														

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Command Table												
R/W#	D/C#	Hex	D7	D6	D5	D4	D ₃	D2	D1	D0	Command	Description
0	0	24	0	0	1	0	0	1	0	0	Write RAM (BW)	After this command, data entries will be written into the BW RAM until another command is written. Address pointers will advance accordingly For Write pixel: Content of Write RAM(BW) = 1 For Black pixel: Content of Write RAM(BW) = 0
0	0	26	0	0	1	0	0	1	1	0	Write RAM (RED)	After this command, data entries will be written into the RED RAM until another command is written. Address pointers will advance accordingly. For Red pixel: Content of Write RAM(RED) = 1 For non-Red pixel [Black or White]: Content of Write RAM(RED) = 0
0	0	27	0	0	1	0	0	1	1	1	Read RAM	After this command, data read on the MCU bus will fetch data from RAM [According to parameter of Register 41h to select reading RAM(BW) / RAM(RED)], until another command is written. Address pointers will advance accordingly. The 1st byte of data read is dummy data.
0	0	28	0	0	1	0	1	0	0	0	VCOM Sense	Enter VCOM sensing conditions and hold for duration defined in 29h before reading VCOM value. The sensed VCOM voltage is stored in register The command required CLKEN=1 and ANALOGEN=1 Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	29	0	0	1	0	1	0	0	1	VCOM Sense Duration	Stabling time between entering VCOM sensing mode and reading acquired. A[6]=1, Normal Mode A[6]=0, Reserve A[3:0] = 09h, duration = 10s. VCOM sense duration = Setting + 1 Seconds
0	1		0	A6	0	0	A ₃	A2	A1	A0		
0	0	2A	0	0	1	0	1	0	1	0	Program VCOM OTP	Program VCOM register into OTP The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.

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Command Table												Command	Description
0	0	34	0	0	1	1	0	1	0	0		CRC calculation	CRC calculation command for OTP content validation. BUSY pad will output high during operation.
0	0	35	0	0	1	1	0	1	0	1		CRC Status Read	CRC Status Read A[15:0] is the CRC readout Value
1	1		A15	A14	A13	A12	A11	A10	A9	A8			
1	1		A7	A6	A5	A4	A3	A2	A1	A0			
0	0	36	0	0	1	1	0	1	1	0		Program selection	OTP Program OTP Selection according to the OTP Selection Control [R37h and R38h] The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	37	0	0	1	1	0	1	1	1		Write Register for Display Option	Write Register for Display Option B[7:0] Display Mode for WS[7:0] C[7:0] Display Mode for WS[15:8] D[7:0] Display Mode for WS[23:16] E[7:0] Display Mode for WS[31:24] F[3:0] Display Mode for WS[35:32] 0: Display Mode 1 1: Display Mode2 F[6]: PingPong for Display Mode 2 F[7]: PingPong for Display Mode 1 1: Ping-Pong 0: Default G[7:0]~J[7:0] module ID /waveform version. Remarks: A[7:0]~J[7:0] can be stored in OTP
0	1		0	0	0	0	0	0	0	0			
0	1		B7	B6	B5	B4	B3	B2	B1	B0			
0	1		C7	C6	C5	C4	C3	C2	C1	C0			
0	1		D7	D6	D5	D4	D3	D2	D1	D0			
0	1		E7	E6	E5	E4	E3	E2	E1	E0			
0	1		F7	F6	F5	F4	F3	F2	F1	F0			
0	1		G7	G6	G5	G4	G3	G2	G1	G0			
0	1		H7	H6	H5	H4	H3	H2	H1	H0			
0	1		I7	I6	I5	I4	I3	I2	I1	I0			
0	1		J7	J6	J5	J4	J3	J2	J1	J0			
0	0	38	0	0	1	1	1	0	0	0		Write Register for User ID	Write Register for User ID A[7:0]~J[7:0]: UserID [10 bytes] Remarks: A[7:0]~J[7:0] can be stored in OTP
0	1		A7	A6	A5	A4	A3	A2	A1	A0			
0	1		B7	B6	B5	B4	B3	B2	B1	B0			
0	1		C7	C6	C5	C4	C3	C2	C1	C0			
0	1		D7	D6	D5	D4	D3	D2	D1	D0			
0	1		E7	E6	E5	E4	E3	E2	E1	E0			
0	1		F7	F6	F5	F4	F3	F2	F1	F0			
0	1		G7	G6	G5	G4	G3	G2	G1	G0			
0	1		H7	H6	H5	H4	H3	H2	H1	H0			
0	1		I7	I6	I5	I4	I3	I2	I1	I0			
0	1		J7	J6	J5	J4	J3	J2	J1	J0			

Command Table													
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description	
0	0	3A	0	0	1	1	1	0	1	0	Reserved	Reserved	
0	0	3B	0	0	1	1	1	0	1	1	Reserved	Reserved	
0	0	3C	0	0	1	1	1	1	0	0	Border Waveform Control	Select border waveform for VBD A[7:0] = C0h [POR], set VBD as HiZ. A [7:6] :Select VBD option	
0	1		A7	A6	A5	A4	0	0	A1	A0		A[7:6]	Select VBD as
												00	GS Transition, Defined in A[1:0]
												01	Fix Level, Defined in A[5:4]
												10	VCOM
												11[POR]	HiZ
												A [5:4] Fix Level Setting for VBD	
												A[5:4]	VBD level
												00[POR]	VSS
												01	VSH1
												10	VSL
												11	VSH2
												A [1:0] GS Transition setting for VBD	
												A[1:0]	VBD Transition
												00[POR]	LUT0
												01	LUT1
												10	LUT2
												11	LUT3
0	0	41	0	1	0	0	0	0	0	1	Read RAM Option	Read RAM Option	
0	1		0	0	0	0	0	0	0	A0		A[0]= 0 [POR] 0: Read RAM corresponding to 24h 1: Read RAM corresponding to 26h	
0	0	44	0	1	0	0	0	1	0	0	Set RAM X - address Start / End position	Specify the start/end positions of the window address in the X direction by an address unit for RAM A[9:0]: XSA[9:0], XStart, POR = 000h B[5:0]: XEA[9:0], XEnd, POR = 3BFh	
0	1		A7	A6	A5	A4	A3	A2	A1	A0			
0	1		-	-	-	-	-	-	A9	A8			
0	1		0	0	B5	B4	B3	B2	B1	B0			
0	1		-	-	-	-	-	-	B9	B8			

Command Table																																																				
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																																								
0	0	45	0	1	0	0	0	1	0	1	Set RAM Y-address Start / End position	Specify the start/end positions of the window address in the Y direction by an address unit for RAM A[8:0]: YSA[8:0], YStart, POR = 000h B[8:0]: YEA[8:0], YEnd, POR = 2A7h																																								
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																										
0	1		-	-	-	-	-	-	A9	A8																																										
0	1		B7	B6	B5	B4	B3	B2	B1	B0																																										
0	1		-	-	-	-	-	-	B9	B8																																										
0	0	46	0	1	0	0	0	1	1	0	Auto Write RED RAM for Regular Pattern	Auto Write RED RAM for Regular Pattern A[7:0] = 00h [POR] A[7]: The 1st step value, POR = 0 A[6:4]: Step Height, POR= 000 Step of alter RAM in Y-direction according to Gate <table><tr><td>A[6:4]</td><td>Height</td><td>A[6:4]</td><td>Height</td></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>960</td></tr></table> A[2:0]: Step Width, POR= 000 Step of alter RAM in X-direction according to Source <table><tr><td>A[2:0]</td><td>Width</td><td>A[2:0]</td><td>Width</td></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>680</td></tr></table> BUSY pad will output high during operation.	A[6:4]	Height	A[6:4]	Height	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	960	A[2:0]	Width	A[2:0]	Width	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	680
A[6:4]	Height	A[6:4]	Height																																																	
000	8	100	128																																																	
001	16	101	256																																																	
010	32	110	512																																																	
011	64	111	960																																																	
A[2:0]	Width	A[2:0]	Width																																																	
000	8	100	128																																																	
001	16	101	256																																																	
010	32	110	512																																																	
011	64	111	680																																																	
0	1		A7	A6	A5	A4	0	A2	A1	A0																																										
0	0	47	0	1	0	0	0	1	1	1	Auto Write B/W RAM for Regular Pattern	Auto Write B/W RAM for Regular Pattern A[7:0] = 00h [POR] Auto Write B/W RAM for Regular Pattern A[7:0] = 00h [POR] <table><tr><td>A[6:4]</td><td>Height</td><td>A[6:4]</td><td>Height</td></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>960</td></tr></table> A[2:0]: Step Width, POR= 000 Step of alter RAM in X-direction according to Source <table><tr><td>A[2:0]</td><td>Width</td><td>A[2:0]</td><td>Width</td></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>680</td></tr></table> uring operation, BUSY pad will output high.	A[6:4]	Height	A[6:4]	Height	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	960	A[2:0]	Width	A[2:0]	Width	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	680
A[6:4]	Height	A[6:4]	Height																																																	
000	8	100	128																																																	
001	16	101	256																																																	
010	32	110	512																																																	
011	64	111	960																																																	
A[2:0]	Width	A[2:0]	Width																																																	
000	8	100	128																																																	
001	16	101	256																																																	
010	32	110	512																																																	
011	64	111	680																																																	
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																										

Command Table												
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	4E	0	1	0	0	1	1	1	0	Set RAM X address counter	Make initial settings for the RAM X address in the address counter (AC) A[9:0]: 000h [POR].
0	1		A7	A6	A5	A4	A3	A2	A1	A0		
0	1		0	0	0	0	0	0	A9	A8		
0	0	4F	0	1	0	0	1	1	1	1	Set RAM Y address counter	Make initial settings for the RAM Y address in the address counter (AC) A[9:0]: 000h [POR].
0	1		A7	A6	A5	A4	A3	A2	A1	A0		
0	1		0	0	0	0	0	0	A9	A8		
0	0	7F	0	1	1	1	1	1	1	1	NOP	This command is an empty command; it does not have any effect on the display module. However, it can be used to terminate Frame Memory Write or Read Commands.

7.Reference Circuit

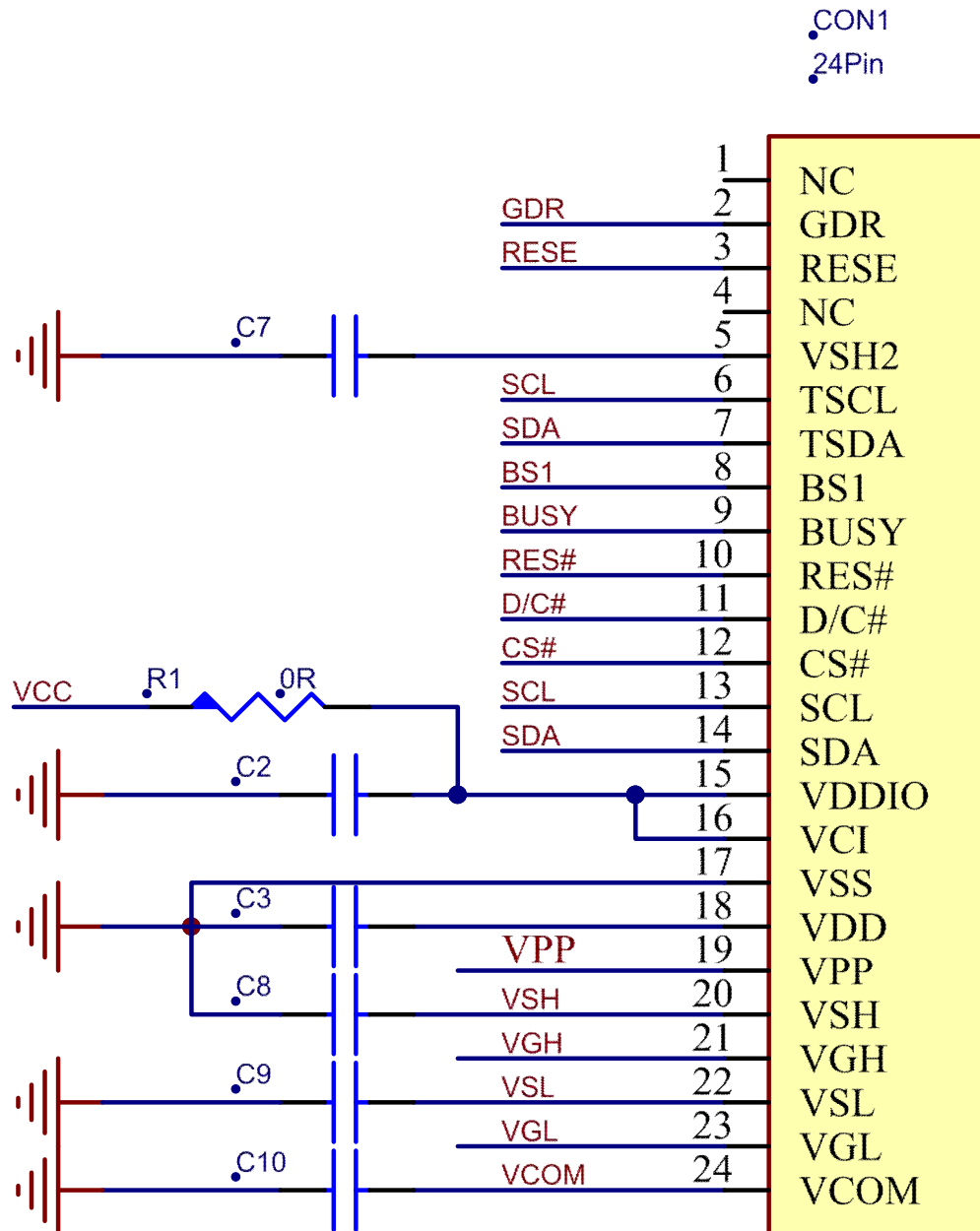


Figure. 7-1

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8 . ABSOLUTE MAXIMUM RATING

Table 8 -1: Maximum Ratings

Symbol	Parameter	Rating	Unit	Humidity	Unit	Note
V _{CI}	Logic supply voltage	-0.5 to +6.0	V	-	-	
T _{OPR}	Operation temperature range	0 to 40	°C	45 to70	%	Normal use is recommended to refresh every 24 hours
-	Transportation temperature range	-25 to 60	°C	-	-	Note10-1
T _{stg}	Storage condition	0 to 40	°C	45 to70	%	Maximum storage time: 5 years
-	After opening the package	0 to 40	°C	45 to70	%	

Note10-1:The transport time is within 10 days for -25°C~0°C or 40°C~60°C

Note 10-2: When the three-color product is stored. The display screen should be kept white and face up. In addition, please be sure to refresh the e-paper every three months.

9 . DC CHARACTERISTICS

The following specifications apply for: VSS=0V, VCI=3.3V, T_{OPR}=25°C.

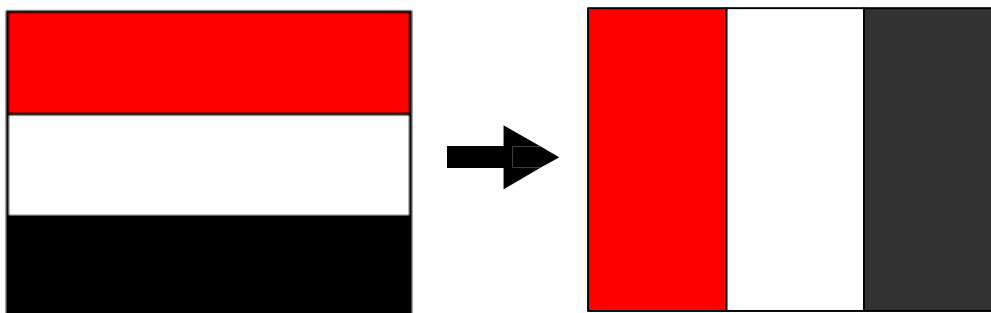
Table 9 -1: DC Characteristics

Symbol	Parameter	Test Condition	Applicable pin	Min.	Typ.	Max.	Unit
V _{CI}	VCI operation voltage	-	VCI	2.5	3	3.7	V
V _{IH}	High level input voltage	-	SDA, SCL, CS#, D/C#,	0.8VDDIO	-	-	V
V _{IL}	Low level input voltage	-	RES#, BS1	-	-	0.2VDDIO	V
V _{OH}	High level output voltage	IOH = -100uA	BUSY,	0.9VDDIO	-	-	V
V _{OL}	Low level output voltage	IOL = 100uA		-	-	0.1VDDIO	V
I _{update}	Module operating current	-	-	-		18	mA
I _{sleep}	Deep sleep mode	VCI=3.3V	-	-		2	uA

- The Typical power consumption is measured using associated 25°C waveform with following pattern transition: from horizontal scan pattern to vertical scan pattern. (Note 12-1)
- The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by XingTai.
- Vcom value will be OTP before in factory or present on the label sticker.

Note 11-1

The Typical power consumption



10. Serial Peripheral Interface Timing

The following specifications apply for: VSS=0V, VCI=2.5V to 3.7V, T_{OPR}=25°C

Write mode

Symbol	Parameter	Min	Typ	Max	Unit
fSCL	SCL frequency (Write Mode)			20	MHz
tCSSU	Time CSB has to be low before the first rising edge of SCLK	20			ns
tCSHLD	Time CSB has to remain low after the last falling edge of SCLK	20			ns
tCSHIGH	Time CSB has to remain high between two transfers	100			ns
tSCLHIGH	Part of the clock period where SCL has to remain high	25			ns
tSCLLOW	Part of the clock period where SCL has to remain low	25			ns
tSISU	Time SI (SDA Write Mode) has to be stable before the next rising edge of SCL	10			ns
tSIHLD	Time SI (SDA Write Mode) has to remain stable after the rising edge of SCL	40			ns

Read mode

Symbol	Parameter	Min	Typ	Max	Unit
fSCL	SCL frequency (Read Mode)			2.5	MHz
tCSSU	Time CSB has to be low before the first rising edge of SCLK	100			ns
tCSHLD	Time CSB has to remain low after the last falling edge of SCLK	50			ns
tCSHIGH	Time CSB has to remain high between two transfers	250			ns
tSCLHIGH	Part of the clock period where SCL has to remain high	180			ns
tSCLLOW	Part of the clock period where SCL has to remain low	180			ns
tSOSU	Time SO (SDA Read Mode) will be stable before the next rising edge of SCL		50		ns
tSOHLD	Time SO (SDA Read Mode) will remain stable after the falling edge of SCL		0		ns

Note: All timings are based on 20% to 80% of VDDIO-VSS

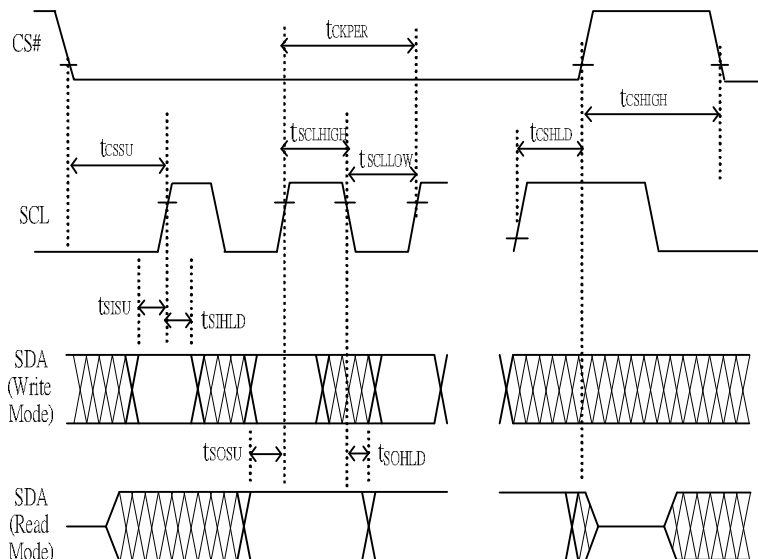


Figure 13-1: Serial peripheral interface characteristics

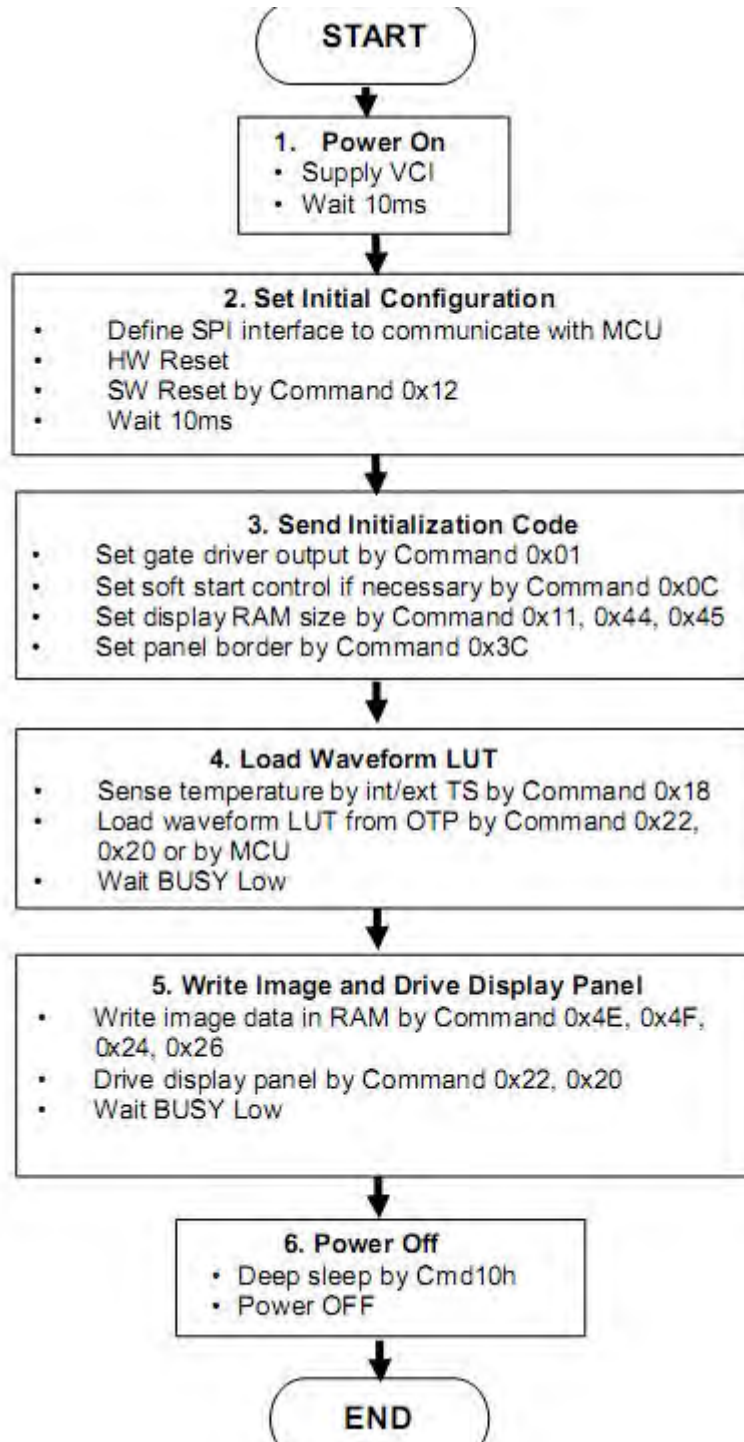
11. Power Consumption

Parameter	Symbol	Conditions	TYP	Max	Unit	Remark
Panel power consumption during update	-	25°C		300	mAs	-
Deep sleep mode	-	25°C		2	uA	-

Mas=update average current × update time

12. Typical Operating Sequence

12.1 Normal Operation Flow



13. Optical characteristics

13.1 Specifications

Measurements are made with that the illumination is under an angle of 45 degrees, the detection is perpendicular unless otherwise specified.

T=25°C

SYMBOL	PARAMETER	CONDITIONS	MIN	TYP.	MAX	UNIT	Note
R	Reflectance	White	30	35	-	%	Note 13-1
Gn	2Grey Level	-	-	$DS+(WS-DS) \times n(m-1)$	-	L*	-
CR	Contrast Ratio	-	10	15	-		-
DS	Dark State L* value		-	13	14		Note 13-1
	Dark State a* value		-	3	5		Note 13-1
WS	White State L* value		63	65	-		Note 13-1
RS	Red State L* value	Red	25	28	-		Note 13-1
	Red State a* value	Red	36	40	-		Note 13-1
Panel's life	-	0°C ~ 40°C		5years	-	-	Note 13-2
Panel	Image Update	Storage and transportation	-	Update the white screen	-	-	-
	Update Time	Operation	-	Suggest Updated once a day	-	-	-

WS : White state, DS : Dark state, RS: Red state

Note 13-1 : Luminance meter : Eye - One Pro Spectrophotometer

Note 13-2 : We don't guarantee 5 years pixels display quality for humidity below 45%RH or above 70%RH;

Suggest Updated once a day;

Note 13-3: To increase the black and white screen clear screen when red has refreshed for a long time, the effect is better.

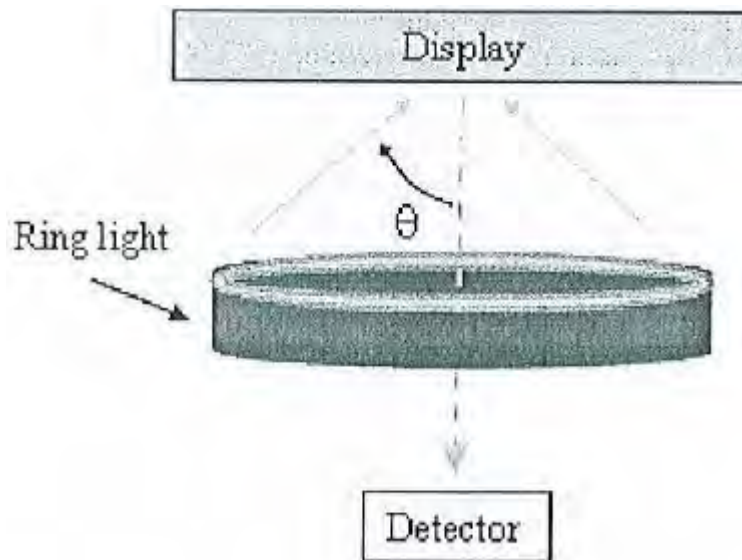
13.2 Definition of contrast ratio

The contrast ratio (CR) is the ratio between the reflectance in a full white area (R1) and the reflectance in a dark area (Rd):

R1: white reflectance

Rd: dark reflectance

$$CR = R1/Rd$$

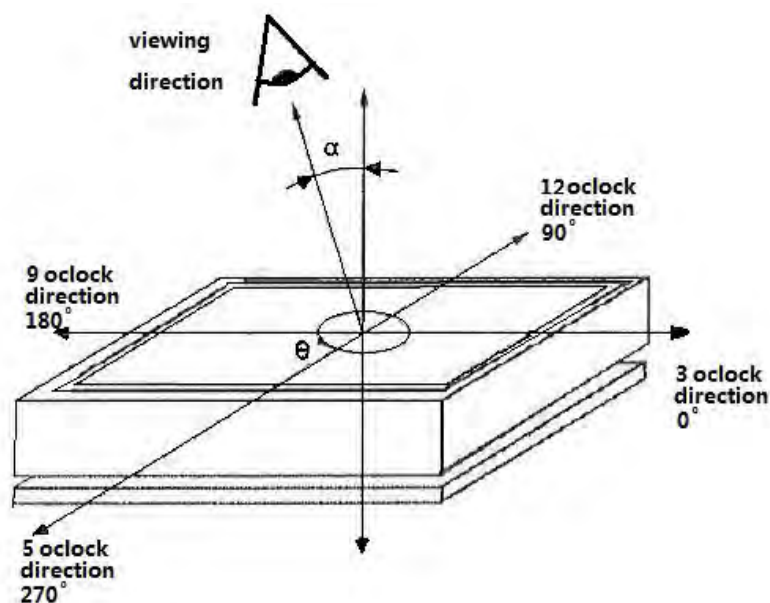


13.3 Reflection Ratio

The reflection ratio is expressed as :

$$R = \text{Reflectance Factor}_{\text{white board}} \times (L_{\text{center}} / L_{\text{white board}})$$

L_{center} is the luminance measured at center in a white area ($R=G=B=1$). $L_{\text{white board}}$ is the luminance of a standard white board. Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.



14. HANDLING, SAFETY AND ENVIRONMENTAL REQUIREMENTS

WARNING
The display glass may break when it is dropped or bumped on a hard surface. Handle with care. Should the display break, do not touch the electrophoretic material. In case of contact with electrophoretic material, wash with water and soap.

CAUTION
The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components.
Disassembling the display module can cause permanent damage and invalidate the warranty agreements.
IPA solvent can only be applied on active area and the back of a glass. For the rest part, it is not allowed.

Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged. Moreover the display is sensitive to static electricity and other rough environmental conditions.

Mounting Precautions	
(1) It's recommended that you consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the module.	
(2) It's recommended that you attach a transparent protective plate to the surface in order to protect the EPD. Transparent protective plate should have sufficient strength in order to resist external force.	
(3) You should adopt radiation structure to satisfy the temperature specification.	
(4) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the PS at high temperature and the latter causes circuit break by electro-chemical reaction.	
(5) Do not touch, push or rub the exposed PS with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of PS for bare hand or greasy cloth. (Some cosmetics deteriorate the PS)	
(6) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach the PS. Do not use acetone, toluene and alcohol because they cause chemical damage to the PS.	
(7) Wipe off saliva or water drops as soon as possible. Their long time contact with PS causes deformations and color fading.	
Product specification	The data sheet contains final product specifications.

Limiting values

Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.
--

Application information

Where application information is given, it is advisory and does not form part of the specification.

Product Environmental certification
--

ROHS

REMARK

All The specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any Post-assembled operation.

15. Reliability test

	TEST	CONDITION	METHOD	REMARK
1	High-Temperature Operation	T=40℃ , RH=35%RH, For 240Hr	IEC 60 068-2-2Bb	
2	Low-Temperature Operation	T = 0℃ for 240 hrs	IEC 60 068-2-2Ab	
3	High-Temperature Storage	T=60℃ RH=35%RH For 240Hr Test in white pattern	IEC 60 068-2-2Bb	
4	Low-Temperature Storage	T = -25℃ for 240 hrs Test in white pattern	IEC 60 068-2-2Ab	
5	High Temperature, High-Humidity Operation	T=40℃ , RH=80%RH, For 168Hr	IEC 60 068-2-3CA	
6	High Temperature, High-Humidity Storage	T=50℃ , RH=80%RH, For 240Hr Test in white pattern	IEC 60 068-2-3CA	
7	Temperature Cycle	-25℃ (30min)~60℃ (30min), 50 Cycle Test in white pattern	IEC 60 068-2-14NB	
8	Package Vibration	1.04G,Frequency : 10~500Hz Direction : X,Y,Z Duration:1hours in each direction	Full packed for shipment	
9	Package Drop Impact	Drop from height of 100 cm on Concrete surface Drop sequence:1 corner, 3edges, 6face One drop for each.	Full packed for shipment	
10	UV exposure Resistance	765 W/m ² for 168hrs,40℃	IEC 60068-2-5 Sa	
11	Electrostatic discharge	Machine model: +/-250V,0Ω,200pF	IEC61000-4-2	

Actual EMC level to be measured on customer application.

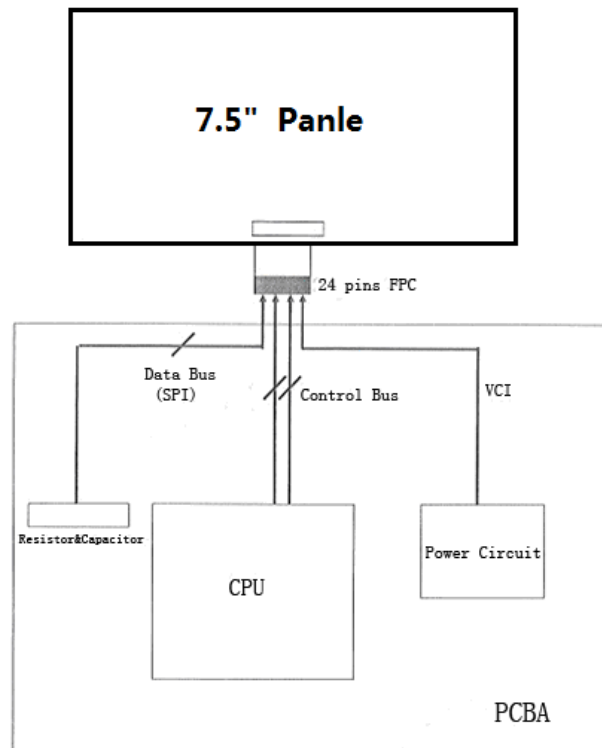
Note1: Stay white pattern for storage and non-operation test.

Note2: Operation is black/white/red pattern , hold time is 150S.

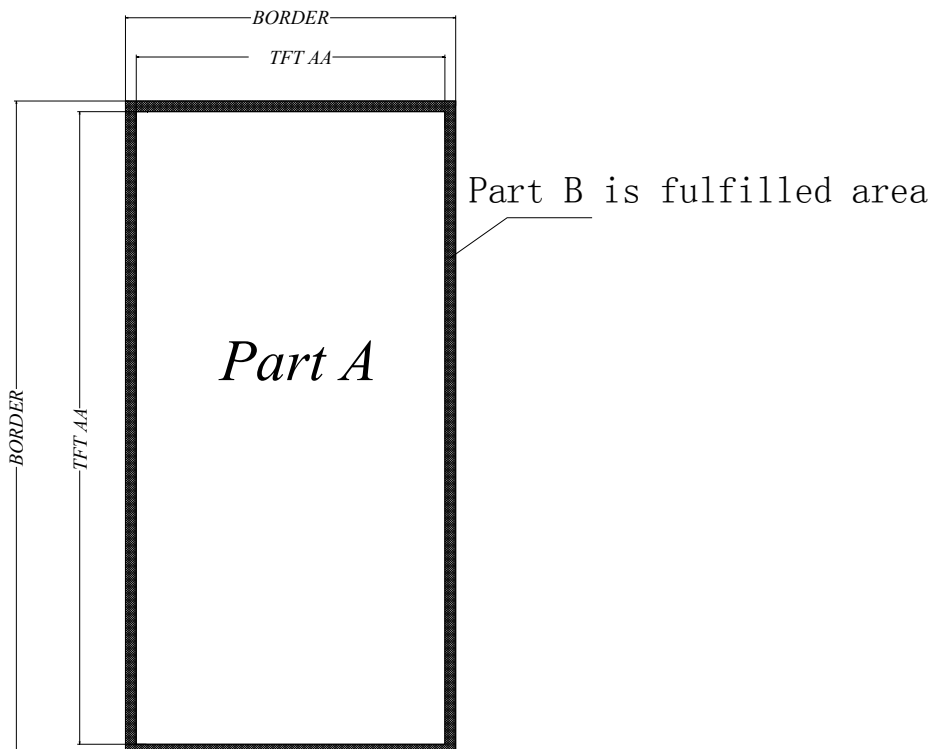
Note3: The function, appearance, opticals should meet the requirements of the test before and after the test.

Note4: Keep testing after 2 hours placing at 20℃-25℃.

16. Block Diagram



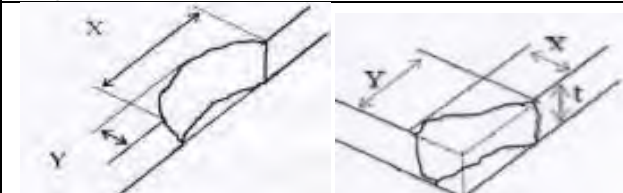
17. PartA/PartB specification

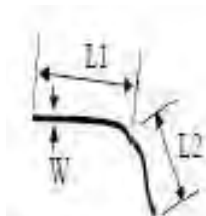


18. Point and line standard

Shipment Inspection Standard

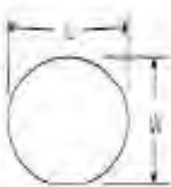
Equipment: Electrical test fixture, Point gauge

Outline dimension	170.2(H) × 111.2(V) × 1.25(D)	Unit: mm	Part-A	Active area	Part-B	Border area
Environment	Temperature	Humidity	Illuminance	Distance	Time	Angle
	19℃～25℃	55%±5%RH	800～1300Lux	300 mm	35Sec	
Defet type	Inspection method	Standard		Part-A		Part-B
Spot	Electric Display	D≤0.2 mm		Ignore		Ignore
		0.2 mm<D≤0.4 mm		N≤4		Ignore
		0.4 mm<D≤0.6 mm		N≤1		Ignore
		D>0.6 mm		Not Allow		Ignore
Display unwork	Electric Display	Not Allow		Not Allow		Ignore
Display error	Electric Display	Not Allow		Not Allow		Ignore
Scratch or line defect(include dirt)	Visual/Film card	L≤2 mm, W≤0.1 mm		Ignore		Ignore
		1.0mm<L≤9.0mm, 0.1<W≤0.2mm,		N≤2		Ignore
		L>9.0 mm, W>0.2 mm		Not Allow		Ignore
PS Bubble	Visual/Film card	D≤0.4mm		Ignore		Ignore
		0.4mm≤D≤0.6mm		N≤4		Ignore
		D>0.6 mm		Not Allow		Ignore
Side Fragment	Visual/Film card	Do not affect the electrode circuit((Corner chipping) X≤8mm, Y≤1mm, Do not affect the electrode circuit, Ignore				
						
Remark	1.Cannot be defect & failure cause by appearance defect;					
	2.Cannot be larger size cause by appearance defect;					
	L=long W=wide D=point size N=Defects NO					



$$L = L1 + L2$$

Line Defect



$$D = (L + W) / 2$$

Spot Defect

L=long W=wide D=point size

19. INSPECTION CRITERIA

19.1 Acceptable Quality Level

Each lot should satisfy the quality level defined as follows

Partition	AQL	Definition
A. Major	0.4%	Functional defective as product
B. Minor	1.5%	Satisfy all functions as product but not satisfy cosmetic standard

19.2 Definition of Lot

One lot means the delivery quantity to customer at one time.

19.3 Condition of Cosmetic Inspection

◆ INSPECTION AND TEST

-FUNCTION TEST

-APPEARANCE INSPECTION

-PACKING SPECIFICATION

◆ INSPECTION CONDITION

- Put under the lamp (20W) at a distance 100mm from

- Tilt upright 45 degree by the front (back) to inspect LCD appearance.

◆ AQL INSPECTION LEVEL

- SAMPLING METHOD: MIL-STD-105D

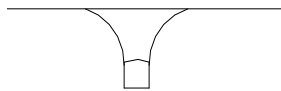
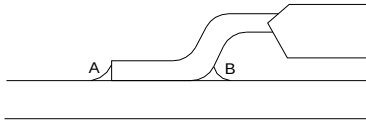
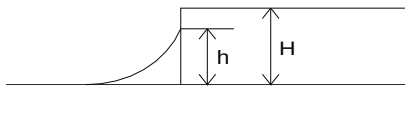
- SAMPLING PLAN: SINGLE

- MAJOR DEFECT: 0.4% (MAJOR)

- MINOR DEFECT: 1.5% (MINOR)

- GENERAL LEVEL: II/NORMAL

19.4 Module Cosmetic Criteria

No.	Item	Judgment Criterion	Partition
1	Difference in Spec.	None allowed	Major
2	Pattern Peeling	No substrate pattern peeling and floating	Major
3	Soldering Defects	No soldering missing	Major
		No soldering bridge	Major
		No cold soldering	Minor
4	Resist Flaw on Substrate	Invisible copper foil(ϕ 0.5mm or more)on substrate pattern	Minor
5	Accretion of Metallic Foreign Matter	No soldering dust	Minor
		No accretion of metallic foreign matters(Not exceed ϕ 0.2mm)	
6	Stain	No stain to spoil cosmetic badly	Minor
7	Plate Discoloring	No plate fading, rusting and discoloring	Minor
8	Solder Amount 1.Lead Parts	a. Soldering side of PCB Solder to form a 'Filet' all around t Solder should not hide the lead form  b.Components side (In case of 'Through Hole PCB') Solder to reach the Components side of PCB 	Minor
	2.Flat Packages	Either 'toe' (A) or 'heel' (B) of the lead to be covered by Filet  Lead form to be assume over solder.	Minor
	3.Chips	$(3/2) H \geq h \geq (1/2)H$ 	Minor

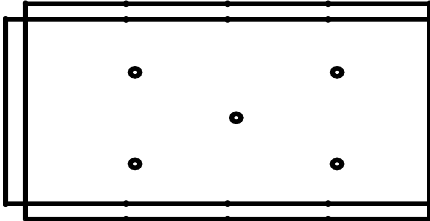
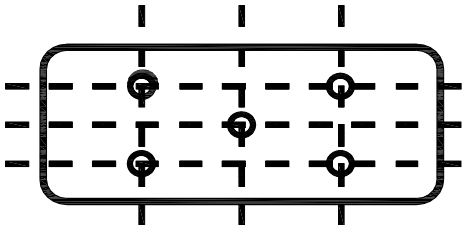
9	Backlight Defects	1.Light fails or flickers.(Major) 2. Color and luminance do not correspond to specifications. (Major) 3.Exceeds standards for display' s blemishes, foreign matter, dark lines or scratches.(Minor)	See list ←
10	PCB Defects	Oxidation or contamination on connectors.* 2. Wrong parts, missing parts, or parts not in specification.* 3.Jumpers set incorrectly.(Minor) 4.Solder(if any)on bezel, LED pad, zebra pad, or screw hole pad is not smooth.(Minor) *Minor if display functions correctly. Major if the display fails.	See list ←
11	Soldering Defects	1. Unmelted solder paste. 2. Cold solder joints, missing solder connections, or oxidation.* 3. Solder bridges causing short circuits.* 4. Residue or solder balls. 5. Solder flux is black or brown. *Minor if display functions correctly. Major if the display fails.	Minor

19.5 Screen Cosmetic Criteria (Non-Operating)

No.	Defect	Judgment Criterion	Partition
1	Spots	In accordance with Screen Cosmetic Criteria (Operating) No.1.	Minor
2	Lines	In accordance with Screen Cosmetic Criteria (Operation) No.2.	Minor
3	Bubbles in Polarizer		Minor
		Size: d mm	
		Acceptable Qty in active area	
		d ≤ 0.3	
		Disregard	
		0.3 < d ≤ 1.0	3
		1.0 < d ≤ 1.5	1
		1.5 < d	0
4	Scratch	In accordance with spots and lines operating cosmetic criteria, When the light reflects on the panel surface, the scratches are not to be remarkable.	Minor
5	Allowable density	Above defects should be separated more than 30mm each other.	Minor
6	Coloration	Not to be noticeable coloration in the viewing area of the LCD panels. Back-lit type should be judged with back-lit on state only.	Minor
7	Contamination	Not to be noticeable.	Minor

19.6 Screen Cosmetic Criteria (Operating)

No.	Defect	Judgment Criterion	Partition	
1	Spots	A) Clear	Minor	
		Size:d mm		Acceptable Qty in active area
		d≤0.1		Disregard
		0.1 < d≤0.2		6
		0.2 < d≤0.3		2
		0.3 < d		0
		Note: Including pin holes and defective dots which must be within one pixel Size. Unclear		
Size:d mm	Acceptable Qty in active area			
d≤0.2	Disregard			
0.2 < d≤0.5	6			
0.5 < d≤0.7	2			
0.7 < d	0			
2	Lines	A) Clear	Minor	
		</		

No.	Defect	Judgment Criterion	Partition
3	Rubbing line	Not to be noticeable.	Minor
4	Allowable density	Above defects should be separated more than 10mm each other.	Minor
5	Rainbow	Not to be noticeable.	Minor
6	Dot size	To be 95%~105% of the dot size (Typ.) in drawing. Partial defects of each dot (ex. pin-hole) should be treated as spot. (see Screen Cosmetic Criteria (Operating) No.1)	Minor
7	Brightness (only back-lit Module)	Brightness Uniformity must be $B_{MAX}/B_{MIN} \leq 2$ - B_{MAX} : Max.value by measure in 5 points - B_{MIN} : Min.value by measure in 5 points Divide active area into 4 vertically and horizontally. Measure 5 points shown in the following figure. 	Minor
8	Contrast Uniformity	Contrast Uniformity must be $B_{MAX}/B_{MIN} \leq 2$ Measure 5 points shown in the following figure. Dashed lines divide active area into 4 vertically and horizontally. Measuring points are located at the inter-sections of dashed line.  Note: B_{MAX} – Max.value by measure in 5 points. B_{MIN} – Min.value by measure in 5 points. O – Measuring points in $\phi 10mm$.	Minor

Note:

(1) Size: $d = (\text{long length} + \text{short length})/2$

(2) The limit samples for each item have priority.

(3) Complexed defects are defined item by item, but if the number of defects is defined in above table, the total number should not exceed 10.

(4) In case of 'concentration', even the spots or the lines of 'disregarded' size should not be allowed. Following three situations should be treated as 'concentration'.

-7 or over defects in circle of $\phi 5mm$.

-10 or over defects in circle of $\phi 10mm$

-20 or over defects in circle of $\phi 20mm$

20. PRECAUTIONS FOR USING

20.1 Handling Precautions

- ◆ This device is susceptible to Electro-Static Discharge (ESD) damage. Observe Anti-Static precautions.
- ◆ EastRising display panel is made of glass. Do not subject it to a mechanical shock by dropping it or impact.
- ◆ If EastRising display panel is damaged and the liquid crystal substance leaks out, be sure not to get any in your mouth. If the substance contacts your skin or clothes, wash it off using soap and water.
- ◆ Do not apply excessive force to the EastRising display surface or the adjoining areas since this may cause the color tone to vary.
- ◆ The polarizer covering the EastRising display surface of the LCD module is soft and easily scratched. Handle this polarizer carefully.
- ◆ If EastRising display surface becomes contaminated, breathe on the surface and gently wipe it with a soft dry cloth. If it is heavily contaminated, moisten cloth with one of the following Isopropyl or alcohol.
- ◆ Solvents other than those above-mentioned may damage the polarizer. Especially, do not use the Water.
- ◆ Exercise care to minimize corrosion of the electrode. Corrosion of the electrodes is accelerated by water droplets, moisture condensation or a current flow in a high-humidity environment.
- ◆ Install the EastRising LCD Module by using the mounting holes. When mounting the LCD module make sure it is free of twisting, warping and distortion. In particular, do not forcibly pull or bend the cable or the backlight cable.
- ◆ Do not attempt to disassemble or process EastRising LCD module.
- ◆ NC terminal should be open. Do not connect anything.
- ◆ If the logic circuit power is off, do not apply the input signals.
- ◆ To prevent destruction of the elements by static electricity, be careful to maintain an optimum work environment.
 - Be sure to ground the body when handling EastRising LCD modules.
 - Tools required for assembling, such as soldering irons, must be properly grounded.
- To reduce the amount of static electricity generated, do not conduct assembling and other work under dry conditions.
- The LCD module is coated with a film to protect the display surface. Exercise care when peeling off this protective film since static electricity may be generated.

20.2 Power Supply Precautions

- ◆ Identify and, at all times, observe absolute maximum ratings for both logic and LC drivers. Note that there is some variance between models.
- ◆ Prevent the application of reverse polarity to VDD and VSS, however briefly.
- ◆ Use a clean power source free from transients. Power-up conditions are occasionally jolting and may exceed the maximum ratings of EastRising modules.
- ◆ The VDD power of EastRising module should also supply the power to all devices that may access the display. Don't allow the data bus to be driven when the logic supply to the module is turned off.

20.3 Operating Precautions

- ♦ DO NOT plug or unplug EastRising module when the system is powered up.
- ♦ Minimize the cable length between EastRising module and host MPU.
- ♦ For models with backlights, do not disable the backlight by interrupting the HV line. Unload inverters produce voltage extremes that may arc within a cable or at the display.
- ♦ Operate EastRising module within the limits of the modules temperature specifications.

20.4 Mechanical/Environmental Precautions

- ♦ Improper soldering is the major cause of module difficulty. Use of flux cleaner is not recommended as they may seep under the electrometric connection and cause display failure.
- ♦ Mount EastRising module so that it is free from torque and mechanical stress.
- ♦ Surface of the LCD panel should not be touched or scratched. The display front surface is an easily scratched, plastic polarizer. Avoid contact and clean only when necessary with soft, absorbent cotton dampened with petroleum benzene.
- ♦ Always employ anti-static procedure while handling EastRising module.
- ♦ Prevent moisture build-up upon the module and observe the environmental constraints for storage tem
- ♦ Do not store in direct sunlight
- ♦ If leakage of the liquid crystal material should occur, avoid contact with this material, particularly ingestion. If the body or clothing becomes contaminated by the liquid crystal material, wash thoroughly with water and soap.

20.5 Storage Precautions

When storing the LCD modules, avoid exposure to direct sunlight or to the light of fluorescent lamps.

Keep EastRising modules in bags (avoid high temperature / high humidity and low temperatures below 0 °C).

Whenever possible, EastRising LCD modules should be stored in the same conditions in which they were shipped from our company.

20.6 Others

Liquid crystals solidify under low temperature (below the storage temperature range) leading to defective orientation or the generation of air bubbles (black or white). Air bubbles may also be generated if the module is subject to a low temperature. If EastRising LCD modules have been operating for a long time showing the same display patterns, the display patterns may remain on the screen as ghost images and a slight contrast irregularity may also appear. A normal operating status can be regained by suspending use for some time. It should be noted that this phenomenon does not adversely affect performance reliability.

To minimize the performance degradation of the LCD modules resulting from destruction caused by static electricity etc., exercise care to avoid holding the following sections when handling the modules.

- Exposed area of the printed circuit board.
- Terminal electrode sections.

21. USING LCD MODULES

21.1 Liquid Crystal Display Modules

EastRising LCD is composed of glass and polarizer. Pay attention to the following items when handling.

- ◆ Please keep the temperature within specified range for use and storage. Polarization degradation, bubble generation or polarizer peel-off may occur with high temperature and high humidity.
- ◆ Do not touch, push or rub the exposed polarizers with anything harder than an HB pencil lead (glass, tweezers, etc.).
- ◆ N-hexane is recommended for cleaning the adhesives used to attach front/rear polarizers and reflectors made of organic substances which will be damaged by chemicals such as acetone, toluene, ethanol and isopropyl alcohol.
- ◆ When EastRising display surface becomes dusty, wipe gently with absorbent cotton or other soft material like chamois soaked in petroleum benzin. Do not scrub hard to avoid damaging the display surface.
- ◆ Wipe off saliva or water drops immediately, contact with water over a long period of time may cause deformation or color fading.
- ◆ Avoid contacting oil and fats.
- ◆ Condensation on the surface and contact with terminals due to cold will damage, stain or dirty the polarizers. After products are tested at low temperature they must be warmed up in a container before coming in contact with room temperature air.
- ◆ Do not put or attach anything on EastRising display area to avoid leaving marks on.
- ◆ Do not touch the display with bare hands. This will stain the display area and degrade insulation between terminals (some cosmetics are determined to the polarizers).
- ◆ As glass is fragile. It tends to become or chipped during handling especially on the edges. Please avoid dropping.

21.2 Installing LCD Modules

- ◆ Cover the surface with a transparent protective plate to protect the polarizer and LC cell.
- ◆ When assembling the LCM into other equipment, the spacer to the bit between the LCM and the fitting plate should have enough height to avoid causing stress to the module surface, refer to the individual specifications for measurements. The measurement tolerance should be $\pm 0.1\text{mm}$.

21.3 Precaution for Handling LCD Modules

Since EastRising LCM has been assembled and adjusted with a high degree of precision; avoid applying excessive shocks to the module or making any alterations or modifications to it.

- ◆ Do not alter, modify or change the shape of the tab on the metal frame.
- ◆ Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached.
- ◆ Do not damage or modify the pattern writing on the printed circuit board.
- ◆ Absolutely do not modify the zebra rubber strip (conductive rubber) or heat seal connector.
- ◆ Except for soldering the interface, do not make any alterations or modifications with a soldering iron.
- ◆ Do not drop, bend or twist EastRising LCM.

21.4 Electro-Static Discharge Control

Since this module uses a CMOS LSI, the same careful attention should be paid to electrostatic discharge as for an ordinary CMOS IC.

- ◆ Make certain that you are grounded when handling LCM.
- ◆ Before remove LCM from its packing case or incorporating it into a set, be sure the module and your body have the same electric potential.
- ◆ When soldering the terminal of LCM, make certain the AC power source for the soldering iron does not leak.
- ◆ When using an electric screwdriver to attach LCM, the screwdriver should be of ground potentiality to minimize as much as possible any transmission of electromagnetic waves produced sparks coming from the commutator of the motor.
- ◆ As far as possible make the electric potential of your work clothes and that of the work bench the ground potential.
- ◆ To reduce the generation of static electricity be careful that the air in the work is not too dried. A relative humidity of 50%-60% is recommended.

21.5 Precaution for Soldering to EastRising LCM

- ◆ Observe the following when soldering lead wire, connector cable and etc. to the LCM.
 - Soldering iron temperature : $280^{\circ}\text{C} \pm 10^{\circ}\text{C}$
 - Soldering time: 3-4 sec.
 - Solder: eutectic solder.

If soldering flux is used, be sure to remove any remaining flux after finishing to soldering operation. (This does not apply in the case of a non-halogen type of flux.) It is recommended that you protect the LCD surface with a cover during soldering to prevent any damage due to flux spatters.

- ◆ When soldering the electroluminescent panel and PC board, the panel and board should not be detached more than three times. This maximum number is determined by the temperature and time conditions mentioned above, though there may be some variance depending on the temperature of the soldering iron.
- ◆ When remove the electroluminescent panel from the PC board, be sure the solder has completely melted, the soldered pad on the PCs board could be damaged.

21.6 Precaution for Operation

- ◆ Driving the EastRising LCD in the voltage above the limit shortens its life.
- ◆ Response time is greatly delayed at temperature below the operating temperature range. However, this does not mean the LCD will be out of the order. It will recover when it returns to the specified temperature range.
- ◆ If EastRising display area is pushed hard during operation, the display will become abnormal. However, it will return to normal if it is turned off and then back on.
- ◆ Condensation on terminals can cause an electrochemical reaction disrupting the terminal circuit. Therefore, it must be used under the relative condition of 40°C , 50% RH.
- ◆ When turning the power on, input each signal after the positive/negative voltage becomes stable.

21.7 Limited Warranty

Unless agreed between EastRising and customer, EastRising will replace or repair any of its LCD modules which are found to be functionally defective when inspected in accordance with EastRising LCD acceptance standards (copies available upon request) for a period of one year from date of shipments. Cosmetic/visual defects must be returned to EastRising within 90 days of shipment. Confirmation of such date shall be based on freight documents. The warranty liability of EastRising limited to repair and/or replacement on the terms set forth above. EastRising will not be responsible for any subsequent or consequential events.

21.8 Return Policy

No warranty can be granted if the precautions stated above have been disregarded. The typical examples of violations are:

- Broken LCD glass.
- PCB eyelet damaged or modified.
- PCB conductors damaged.
- Circuit modified in any way, including addition of components.
- PCB tampered with by grinding, engraving or painting varnish.
- Soldering to or modifying the bezel in any manner.

Module repairs will be invoiced to the customer upon mutual agreement. Modules must be returned with sufficient description of the failures or defects. Any connectors or cable installed by the customer must be removed completely without damaging the PCB eyelet's, conductors and terminals.

22. IMAGE STICKING

22.1 What is Image Sticking?

If you remain a fixed image on LCD Display for a long period of time, you may experience a phenomenon called Image Sticking. Image Sticking - sometimes also called "image retention" or "ghosting" - is a phenomenon where a faint outline of a previously displayed image remains visible on the screen when the image is changed. It can occur at variable levels of intensity depending on the specific image makeup, as well as the amount of time the core image elements are allowed to remain unchanged on the screen.

22.2 What Causes Image Sticking and How to Avoid?

1. The e-Paper display cannot be powered on for a long time, you must set e-Paper display to sleep mode or power off when it needn't refresh ,otherwise e-Paper keeps in high voltage status for long time which will damage e-Paper and cannot be fixed. We suggest customers to update e-Paper display every 24 hours or at least 10 days to update again. Otherwise, ghost of the last content may cannot be cleared.

It is also recommended that customer ships or stores the e-Paper display with completely white image to avoid image sticking issue and refresh

2. Three-color e-Paper display is normal to be a little "color" . You can refresh it to white to keep it upward for storage.

3. The e-Paper display ignores the data sent when it is in sleep mode, you need to initialize it for properly refreshing. The e-Paper display cannot refresh directly under sunlight. The refresh steps should be done indoor.

4. For those e-Paper displays which support partial refresh, you cannot use partial refresh all the time. A full refresh should be done to clear screen after several times (partial refresh), otherwise, e-Paper display will be damaged and cannot fixed.

23. STORAGE

We recommend customers to refresh three-color e-Paper displays one by one if storage period is more than half a year, otherwise the image on display may be unclear as below image shows.

